C1.6 Concepts required for the ACE specification

The ACE specification defines the following concepts:

- *Domains*
- *Barriers* on page C1-148
- *Distributed Virtual Memory* on page C1-149.

C1.6.1 Domains

The ACE protocol uses a concept called shareability domains. A shareability domain is a set of master components that enables a master component to determine which other master components to include when issuing coherency or barrier transactions.

For coherent transactions, a master component uses the shareability domain to determine which other master components might have a copy of the addressed location in their local cache. The interconnect component uses this information to determine, for any given transaction, which other master components must be snooped to complete the transaction.

For barrier transactions, a master component uses the shareability domain to determine which other master components the barrier is establishing an ordering relationship with. The domain of a barrier transaction can be used to determine how far a barrier transaction must propagate, and the blocking properties necessary to establish the required ordering.

The ACE protocol defines the following levels of shareability domain:

Non-shareable The domain contains a single master component.

Inner Shareable The domain can include additional master components.

Outer Shareable The domain contains at least all master components in the Inner domain, but can include additional master components.

System The domain includes all master components in the system.

Figure C1-3 shows an example set of shareability domains for a system that includes master components 0-9.

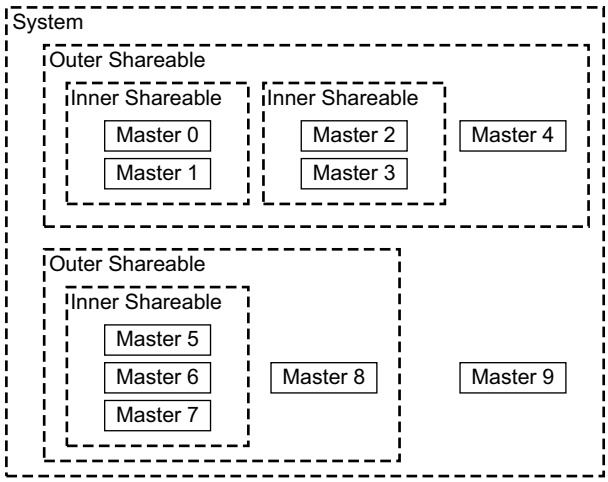


Figure C1-3 Shareability domains

Although multiple Non-shareable, Inner Shareable, and Outer Shareable domains can exist in a system, there must be a single consistent definition of the master components that are contained in each domain. For example, in Figure C1-3, because master 0 has master 1 in its Inner Shareable domain, then master 1 must have master 0 in its Inner Shareable domain.

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Domains are defined as non-overlapping. For each master component in an Outer Shareable domain, all the other master components, in the Inner Shareable domain that includes that master component, must also be included in the same Outer Shareable domain.

For transactions that must be visible to all other master components in the system, the System domain is used. Because System domain transactions include master components that do not have hardware-coherent caches, these transactions must not be cached at any level.

In Figure C1-4, from the perspective of master 1:

- the cache of master 1 is a local cache
- the caches of masters 2-6 are peer caches
- caches of masters 1-3 are in the Inner Shareable domain
- caches of masters 1-6 are in the Outer Shareable domain
- cache 1 is a downstream cache.

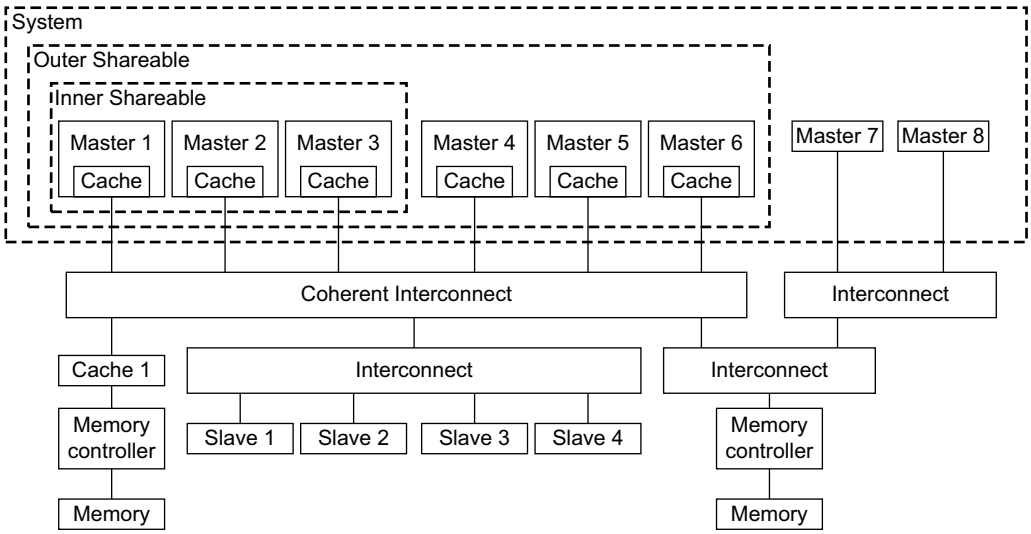


Figure C1-4 Example system using shareability domains

C1.6.2 Barriers

Barrier transactions provide guarantees about the ordering and observation of transactions in a system. The following types of barrier transaction are supported:

Memory barriers

A master component issues a memory barrier to guarantee that if another master component in the appropriate domain can observe any transaction after the barrier it must be able to observe every transaction prior to the barrier.

Synchronization barriers

A master component issues a synchronization barrier to determine when every master component in the appropriate domain can observe all transactions that preceded the barrier transaction. For System domain synchronization barriers, all transactions issued before the barrier transaction must have reached the destination slave components before the barrier transaction completes.

A barrier transaction has an address phase and a response, but no data transfer occurs. A master component must issue a barrier transaction on both the read address channel and the write address channel.

Barriers can enforce ordering because a master component must not issue any read or write transaction that must be ordered after the barrier until the master component has received a response for the barrier on both read data and write response channels.

See Chapter C8 *Barrier Transactions* for more information.

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C1.6.3 Distributed Virtual Memory

ACE supports *Distributed Virtual Memory* (DVM) and includes transactions that permit virtual memory system management. Figure C1-5 shows the basic parts of a virtual memory system.

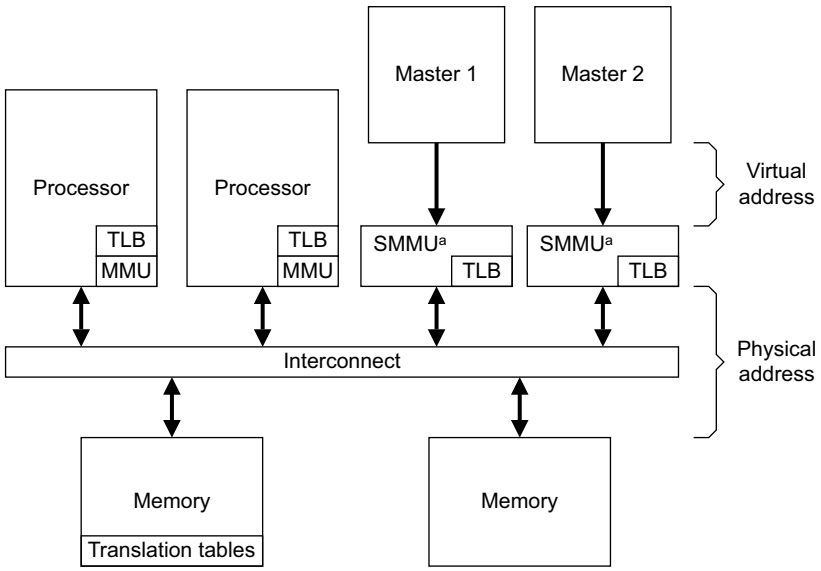


Figure C1-5 Virtual memory system

In Figure C1-5, the *System Memory Management Units* (SMMUs) translate addresses in the virtual address space to addresses in the physical address space. Although all components in the system must use a single physical address space, SMMU components enable different master components to operate in their own independent virtual address or intermediate physical address space.

A typical process in the virtual memory system shown in Figure C1-5 might operate as follows:

1. A master component operating in a *virtual address* (VA) space issues a transaction that uses a VA.
2. The SMMU receives the VA, for translation to a *physical address* (PA):
   - if the SMMU has recently performed the requested translation then it might obtain a cached copy of the translation from its TLB
   - otherwise, the SMMU must perform a *translation table walk*, accessing translation table in memory to obtain the required VA to PA translation.
3. The SMMU uses the PA to issue the transaction for the requesting component.

At step 2 of this process, the translation for the required VA might not exist. In this case, the translation table walk generates a fault, that must be notified to the agent that maintains the translation tables. For the required access to proceed, that agent must then provide the required VA to PA translation. Typically, it will update the translation tables with the required information.

Maintaining the translation tables can require changes to translation table entries that are cached in TLBs. To prevent the use of these entries, a DVM message can be used to issue a TLB invalidate operation.

When the translation tables have been updated, and the required TLB invalidations performed, a DVM Sync transaction is used to ensure that all required transactions have completed.

See Chapter C12 *Distributed Virtual Memory Transactions* for more information.

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